

Binding Potentials and Interaction Gates between Microwave-Dressed Rydberg Atoms

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We demonstrate finite range binding potentials between pairs of Rydberg atoms interacting with each other via attractive and repulsive van der Waals potentials and driven by a microwave field. We show that, using destructive quantum interference to cancel single-atom Rydberg excitation, the Rydberg-dimer states can be selectively and coherently populated from the two-atom ground state. This can be used to realize a two-qubit interaction gate which is not susceptible to mechanical forces between the atoms and is therefore immune to motional decoherence.

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Atoms in high-lying Rydberg states exhibit many remarkable features, including long lifetimes and giant polarizability [1]. The resulting strong, long-range, resonant (Förster) and nonresonant (van der Waals) dipole-dipole (DD) interactions between the atoms can suppress multiple Rydberg excitations within a certain blockade distance [2–5]. In combination with laser and microwave field manipulation of atomic states, these interactions form the basis for quantum information processing [4] with individual atoms [2,6–9] and atomic ensembles [3,4,10]. Furthermore, cold atoms excited to Rydberg states represent a flexible platform to simulate [11] and study few-body [12–21] and many-body physics [22–39].

A paradigmatic interaction phenomenon is the formation of a bound pair of particles. An electron in a Rydberg orbit scattering off a ground-state atom can sustain a weakly bound molecule with a permanent dipole moment [12,13]. Macrodimers of two Rydberg atoms can form due to van der Waals (VDW) interactions in a static electric field [14,15] or through the mixing of orbital angular momentum states [16]. Another binding mechanism relies on the Stark-shifted DD interactions which can support Rydberg dimers [17] and trimers [18].

Here we identify a new mechanism to obtain long-range binding potentials between Rydberg atoms. Our two-atom potential curves result from microwave field coupling between a pair of Rydberg states of each atom [19]. Atoms in these states interact with each other via repulsive and attractive VDW potentials, and via typically weaker DD exchange interaction on the allowed microwave transition. When the attractive VDW interaction is comparable to, or stronger than, the repulsive VDW (and DD) interaction, the microwave-dressed potential energy curves have pronounced wells located in the vicinity of crossings of the two-atom attractive and repulsive potentials, in the frame rotating with the microwave frequency. The microwave field, which is detuned from the transition resonance of a

single atom, lifts the degeneracy and causes level anti-crossing in the two-atom basis. We note that the combination of resonant and nonresonant DD interactions between the Rydberg atoms can also result in a binding potential [17].

We next address the question of how to selectively populate these Rydberg-dimer states starting from the two-atom ground state. The strong microwave field induces a broad dark resonance [40] for the laser excitation of a single (non-interacting) atom. We show that within this electromagnetically induced transparency (EIT) [41] window, a smooth probe laser pulse populates only the two-atom Rydberg manifold. We then propose to employ such a Rydberg-dimer state to realize the universal CPHASE quantum logic gate between a pair of qubits represented by atoms trapped at a suitable distance from each other. This is implemented by coherent excitation and deexcitation of the atoms with a probe pulse of effective area 2π . The quantum interference responsible for EIT prevents Rydberg excitation of a single atom, and only an appropriate two-atom state acquires a conditional phase π . Since during the gate operation we populate the two-atom Rydberg state at the bottom of a potential well, there is no mechanical force between the atoms and motional decoherence [21] is suppressed.

Consider a pair of atoms with the Rydberg states $|e\rangle$ and $|r\rangle$ coupled by a microwave field with Rabi frequency Ω and detuning Δ ; see Fig. 1(a). In the frame rotating with the microwave field frequency, the interaction Hamiltonian for atom $j = 1, 2$ reads $\mathcal{V}^j = -\hbar\Delta\hat{\sigma}_{rr}^j - \hbar\Omega(\hat{\sigma}_{re}^j + \hat{\sigma}_{er}^j)$, where $\hat{\sigma}_{\alpha\beta}^j \equiv |\alpha_j\rangle\langle\beta_j|$ denote the atomic operators. Atoms in states $|e\rangle$ and $|r\rangle$ interact via the VDW potentials $\mathcal{W}_{ee} = \hbar(C_6^{ee}/R^6)\hat{\sigma}_{ee}^1 \otimes \hat{\sigma}_{ee}^2$ and $\mathcal{W}_{rr} = \hbar(C_6^{rr}/R^6)\hat{\sigma}_{rr}^1 \otimes \hat{\sigma}_{rr}^2$, where R is the interatomic distance and C_6^{ee} and C_6^{rr} are the corresponding VDW coefficients. There is also a resonant DD (exchange) interaction between the atoms

$\mathcal{D}_{er} = \hbar \sum_{i \neq j} (C_3^{er}/R^3) \hat{\sigma}_{re}^i \otimes \hat{\sigma}_{er}^j$, and an effective VDW interaction $\mathcal{W}_{er} = \hbar \sum_{i \neq j} (C_6^{er}/R^6) \hat{\sigma}_{rr}^i \otimes \hat{\sigma}_{ee}^j$ which arises from nonresonant DD interaction with shifted Rydberg level(s) [42]. The total Hamiltonian for the pair of Rydberg atoms is $\mathcal{H}_{2\text{Ry}} = \mathcal{V}^1 + \mathcal{V}^2 + \mathcal{W}_{ee} + \mathcal{W}_{rr} + \mathcal{D}_{er} + \mathcal{W}_{er}$. Below we focus on the case of large detuning $|\Delta| \gg \Omega$, which exhibits rich structure and deep potential wells; the (near)-resonant case $|\Delta| \lesssim \Omega$ is discussed in [42].

Our two-atom basis set consists of states $|ee\rangle \equiv |e_1 e_2\rangle$, $|er_{\pm}\rangle \equiv 1/\sqrt{2}(|e_1 r_2\rangle \pm |r_1 e_2\rangle)$, and $|rr\rangle \equiv |r_1 r_2\rangle$ [Fig. 1(b)]. When $\Omega \rightarrow 0$, the (bare) energies of states $|ee\rangle$ and $|rr\rangle$ are given by $E_{ee}/\hbar = C_6^{ee}/R^6$ and $E_{rr}/\hbar = -2\Delta + C_6^{rr}/R^6$, with $C_6^{ee} > 0$ (repulsive VDW interaction) and $C_6^{rr} < 0$ (attractive VDW interaction) while $\Delta < 0$ (red detuning). In turn, the resonant DD interaction lifts the degeneracy of states $|er_{\pm}\rangle$ whose energies are $E_{er_{\pm}}/\hbar = -\Delta \pm C_3^{er}/R^3 + C_6^{er}/R^6$, where $C_6^{er} > 0$ is due to the effective VDW interaction [42]. Figure 1(c) shows the dependence of bare energy levels $E_{\alpha\beta}$ on R . For vanishing VDW and DD interactions, $R \rightarrow \infty$, $E_{ee} < E_{er_{\pm}} < E_{rr}$, while in the opposite limit of very strong (compared to $|\Delta|$) interactions, $R \rightarrow 0$, $E_{ee} > E_{er_{+}} > E_{er_{-}} > E_{rr}$ (in the rotating frame). There are three level-crossing points of interest: $E_{ee} = E_{er_{+}} \equiv E_{c1}$ at R_1 , $E_{ee} = E_{rr} \equiv E_{c2}$ at R_2 , and $E_{rr} = E_{er_{+}} \equiv E_{c3}$ at R_3 ; see [42].

Consider first the antisymmetric state $|er_{-}\rangle$. At a large distance, the potential $E_{er_{-}} \propto -R^{-3}$ is attractive due to the long-range resonant DD interaction, but at smaller distances, the repulsive VDW interaction dominates, $E_{er_{-}} \propto R^{-6}$. Hence, $E_{er_{-}}$ has a potential well, $\partial_R E_{er_{-}} = 0$, around $R_- = \sqrt[3]{2C_6^{er}/C_3^{er}}$ where the VDW repulsion overcomes the DD attraction [Fig. 1(c)]. This simplified treatment captures the essential physics of the binding potential for DD interacting atoms presented in [17]. The antisymmetric state $|er_{-}\rangle$ will not play a role in our subsequent analysis since, for not too large dephasing, it is decoupled from the other two-atom states and the microwave field, even when $\Omega \neq 0$.

We are thus left with three basis states $|ee\rangle$, $|er_{+}\rangle$ and $|rr\rangle$ coupled sequentially by the microwave field with rate $\sqrt{2}\Omega$; see Fig. 1(b). At large distances $R > R_{2,3}$, the VDW (and DD) interactions are much weaker than $|\Delta|$ and the red-detuned ($\Delta < 0$) microwave field induces ac Stark shifts $\pm(2|\Omega|^2/\Delta)$ of levels $|ee\rangle$ and $|rr\rangle$ [Fig. 1(c)]. At small distances, $R < R_{1,2}$, the VDW and DD shifts are so large that levels $|ee\rangle$, $|rr\rangle$, and $|er_{+}\rangle$ decouple from the field. At the bare energy level crossing points R_1 and R_3 , the microwave field becomes resonant with the transitions $|er_{+}\rangle \leftrightarrow |ee\rangle$ and $|er_{+}\rangle \leftrightarrow |rr\rangle$. This leads to avoided crossings of the microwave-dressed energy levels E_u and E_m which are repelled from $E_{c1,c3}$ by $\pm\sqrt{2}\Omega$. The upper potential curve E_u has now a broad well near the crossing point R_3 of the bare levels $E_{er_{+}}$ and E_{rr} determined by the

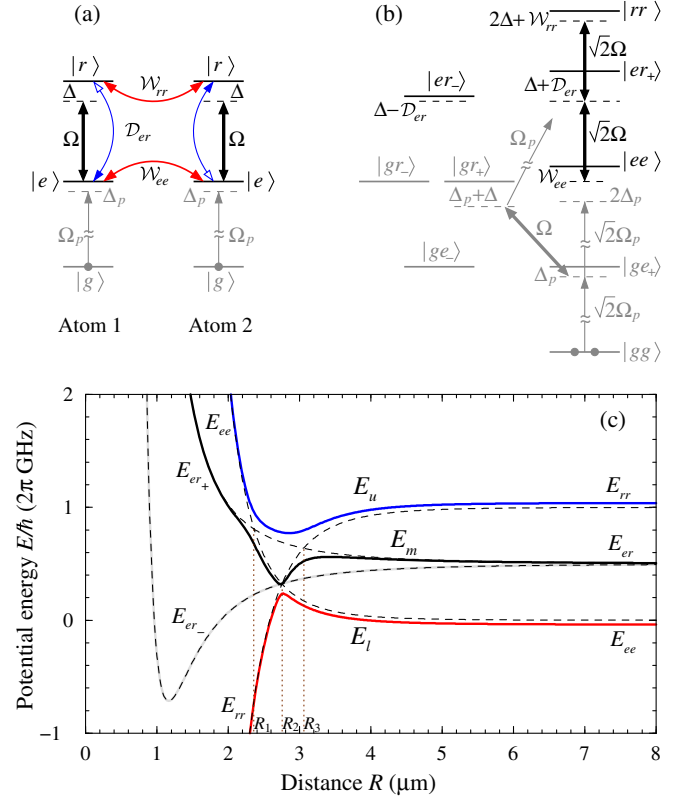


FIG. 1 (color online). (a) Atoms 1 and 2 with Rydberg states $|e\rangle$ and $|r\rangle$ interact via repulsive $\mathcal{W}_{ee}$ and attractive $\mathcal{W}_{rr}$ VDW interactions and DD (exchange) interaction $\mathcal{D}_{er}$, while a microwave field drives transition $|e\rangle \leftrightarrow |r\rangle$ with Rabi frequency Ω and detuning Δ . The Rydberg states can be excited from the ground state $|g\rangle$ by a probe laser field with Rabi frequency Ω_p and detuning Δ_p . (b) Diagram of levels and couplings in the two-atom basis. Antisymmetric states $|ge_{-}\rangle$, $|gr_{-}\rangle$, and $|er_{-}\rangle$ are decoupled from the laser and microwave fields. (c) Potential curves E_l , E_m , E_u for the two-atom microwave-dressed Rydberg states of Rb atoms with $|e\rangle \equiv |60S_{1/2}, m_j = +\frac{1}{2}\rangle$, $|r\rangle \equiv |60P_{3/2}, m_j = +\frac{3}{2}\rangle$, $\theta = \pi/2$, $\Omega = 2\pi \times 0.1$ GHz, and $\Delta = -5\Omega$. Dashed curves correspond to bare ($\Omega \rightarrow 0$) energy levels E_{ee} , $E_{er_{\pm}}$, E_{rr} crossing at $R_{1,2,3}$.

weakly repulsive DD interaction and strongly attractive VDW interaction. For the parameters used in Fig. 1(c), at the bottom of the potential well the vibration frequency of the two-atom relative motion is $\nu_u \approx 2\pi \times 450$ kHz [42].

Similarly, at the bare level crossing point R_2 , the microwave field couples states $|ee\rangle \leftrightarrow |rr\rangle$, via nonresonant intermediate state $|er_{+}\rangle$, with the two-photon Rabi frequency $\Omega^{(2)} = \hbar\sqrt{2}\Omega^2/(E_{c2} - E_{er_{+}}(R_2))$. Now the microwave-dressed energy levels E_l and E_m are repelled from E_{c2} by $\pm\Omega^{(2)}$, with the result that the middle potential curve E_m has a narrow well with a minimum near E_{c2} . With the above parameters, $\Omega^{(2)} = 2\pi \times 55$ MHz and the two-atom relative vibrational frequency in the vicinity of the potential well minimum R_m is $\nu_m \approx 2\pi \times 2$ MHz [42]. Note that the interatomic potentials of Fig. 1(c) are

azimuthally symmetric and robust with respect to small variations of the angle θ between the quantization axis and the two-atom separation vector, as shown in [42].

We next consider the excitation of the Rydberg states of atoms from the ground state $|g\rangle$ by a laser field acting on transition $|g\rangle \rightarrow |e\rangle$ with the Rabi frequency Ω_p and detuning Δ_p ; see Fig. 1(a). The total Hamiltonian for the pair of atoms is now $\mathcal{H} = \mathcal{H}_{2\text{Ry}} + \mathcal{V}_p^1 + \mathcal{V}_p^2$ with the interaction Hamiltonian $\mathcal{V}_p^j = \hbar\Delta_p\hat{\sigma}_{gg}^j - \hbar\Omega_p(\hat{\sigma}_{eg}^j + \hat{\sigma}_{ge}^j)$. We simulate the dynamics of the system using the master equation for its density operator $\hat{\rho}$, $\partial_t\hat{\rho} = -(i/\hbar)[\mathcal{H}, \hat{\rho}] + \mathcal{L}\hat{\rho}$, where the Liouvillian $\mathcal{L}\hat{\rho} = \sum_{j=1,2}(\mathcal{L}_g^j\hat{\rho} + \mathcal{L}_e^j\hat{\rho} + \mathcal{L}_r^j\hat{\rho})$, with $\mathcal{L}_a^j\hat{\rho} = \frac{1}{2}[2\hat{L}_a^j\hat{\rho}\hat{L}_a^{j\dagger} - \{\hat{L}_a^{j\dagger}\hat{L}_a^j, \hat{\rho}\}]$, accounts for the relaxation processes affecting the atoms [46,47]. These include the slow population decay of Rydberg states $|e\rangle, |r\rangle$ with rates $\Gamma_{e,r}$, and the usually more rapid decay of atomic coherences $\hat{\sigma}_{eg}$ and $\hat{\sigma}_{rg}$ with the total rate $\gamma = \gamma_g + \frac{1}{2}\Gamma_{e,r}$ ($\gamma_g \gg \Gamma_{e,r}$) which originates from the laser phase fluctuations, Doppler shifts due to thermal atomic motion, and intermediate state decay when $|g\rangle \rightarrow |e\rangle$ is a two-photon transition [6–8,10,38,39]. The corresponding Lindblad generators are $\hat{L}_e^j = \sqrt{\Gamma_e}\hat{\sigma}_{ge}^j$, $\hat{L}_r^j = \sqrt{\Gamma_r}\hat{\sigma}_{gr}^j$, and $\hat{L}_g^j = \sqrt{\gamma_g/2}(\hat{\sigma}_{gg}^j - \hat{\sigma}_{ee}^j - \hat{\sigma}_{rr}^j)$.

We assume that a short probe laser pulse $\Omega_p(t)$ irradiates the pair of free (thermal) atoms, followed by the detection of atoms in the Rydberg states, e.g., through dc field ionization. We vary the probe field frequency Δ_p and interatomic distance R from pulse to pulse to obtain the Rydberg excitation probabilities shown in Fig. 2. The probabilities of single $P_{1\text{Ry}} = \text{tr}(\hat{\rho}\hat{\Pi}_{1\text{Ry}})$ and double $P_{2\text{Ry}} = \text{tr}(\hat{\rho}\hat{\Pi}_{2\text{Ry}})$ excitations are defined through the projectors $\hat{\Pi}_{1\text{Ry}} \equiv \sum_{i \neq j}(\hat{\sigma}_{ee}^i + \hat{\sigma}_{rr}^i) \otimes \hat{\sigma}_{gg}^j$ and $\hat{\Pi}_{2\text{Ry}} \equiv (\hat{\sigma}_{ee}^1 + \hat{\sigma}_{rr}^1) \otimes (\hat{\sigma}_{ee}^2 + \hat{\sigma}_{rr}^2)$, where $\hat{\sigma}_{gg}^j + \hat{\sigma}_{ee}^j + \hat{\sigma}_{rr}^j = \mathbb{1}_j$. An actual experiment may or may not resolve the Rydberg state and atom (or ion) number; therefore, both $P_{1\text{Ry}}$ and $P_{2\text{Ry}}$ are treated on equal footing.

Clearly, single-atom excitation $P_{1\text{Ry}}$ or pair excitation $P_{2\text{Ry}}$ at large interatomic distances R are unaffected by the Rydberg-state interactions. Scanning the frequency of the laser field, we then probe the microwave field induced Autler-Townes doublet $\lambda_{\pm} = -\frac{1}{2}\Delta \pm \sqrt{\frac{1}{4}\Delta^2 + \Omega^2}$ of the Rydberg states $|e\rangle$ and $|r\rangle$. For large microwave detuning $|\Delta| \gg \Omega$, the probe field resonances are at $\Delta_p^{(-)} = \lambda_- \approx -\Omega^2/|\Delta|$ and $\Delta_p^{(+)} = \lambda_+ \approx |\Delta| + \Omega^2/|\Delta|$, as expected [cf. Fig. 1(c)]. Importantly, in the frequency region between $\Delta_p^{(-)}$ and $\Delta_p^{(+)}$, destructive quantum interference leads to a dark resonance (EIT window) for the probe pulse [40,41]: If the pulse envelope varies adiabatically, $\partial_t\Omega_p < \Omega_p|\lambda_+ - \lambda_-|$, it does not excite the bright eigenstates of the three-level atom, which correspond to the

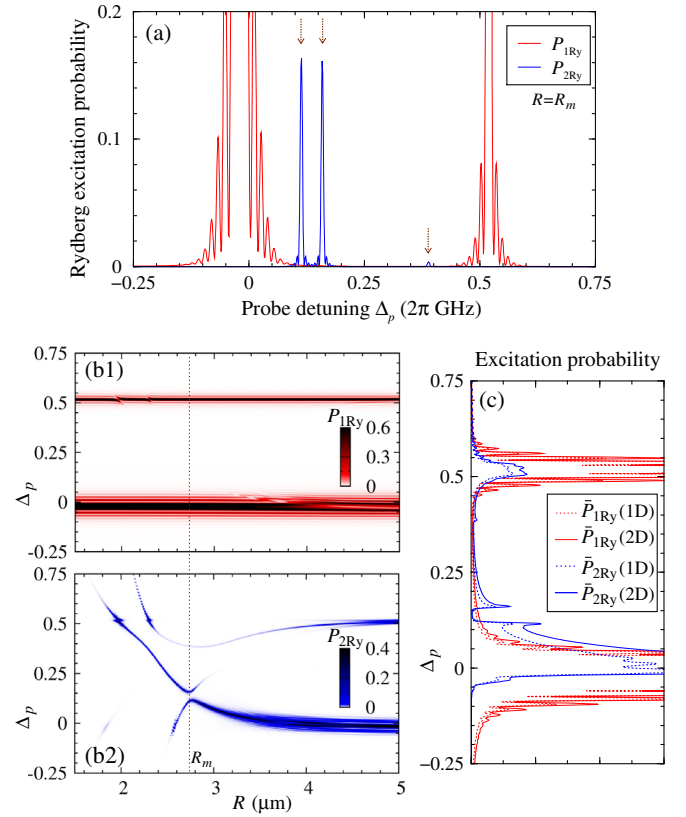


FIG. 2 (color online). Rydberg excitation spectra for two atoms excited from the ground state $|gg\rangle$ by a smooth probe pulse $\Omega_p(t)$. (a) One- and two-atom excitation probabilities $P_{1\text{Ry}}$ and $P_{2\text{Ry}}$ at interatomic distance $R_m = 2.74 \mu\text{m}$. (b1), (b2) Density plots of $P_{1\text{Ry}}, P_{2\text{Ry}}$ versus distance R [vertical dotted line marks $R = R_m$, cf. (a)]. (c) Spatially averaged excitation probabilities $\bar{P}_{1\text{Ry}}, \bar{P}_{2\text{Ry}}$ for pairs of atoms in a 1D or 2D volume of linear dimension $L = 5 \mu\text{m}$. In the simulations, the decay rates are $\Gamma_{e,r} = 5 \text{ kHz}$ and $\gamma_g = 2\pi \times 100 \text{ kHz}$, the probe Rabi frequency $\Omega_p = 2\pi \times 10 \text{ MHz}$ (0.1Ω), and duration $\tau_p = 80 \text{ ns}$ (flat-top pulse with Gaussian leading and trailing edges of 10 ns duration); other parameters are as in Fig. 1(c).

Autler-Townes doublet when $\Omega_p \ll \Omega$. During the interaction with the pulse, the populations of the Rydberg states are $\langle\hat{\sigma}_{ee}\rangle \approx 0$ and $\langle\hat{\sigma}_{rr}\rangle \approx \Omega_p^2/\Omega^2$, but after the interaction, $\Omega_p \rightarrow 0$, the atom returns to the ground state with ideally unit probability $\langle\hat{\sigma}_{gg}\rangle \approx 1$. Atomic coherence relaxation γ , however, reduces the transparency causing residual population of the excited states.

Hence, within the EIT window for a single (noninteracting) atom, we can probe the two-atom Rydberg resonances $E_{l,m,u}$. In Fig. 2(a) we show the Rydberg excitation probabilities versus probe detuning Δ_p for the atoms at distance $R = R_m$ of the E_m potential minimum. Figure 2(b) summarizes the results of our simulations for all (relevant) probe frequencies and interatomic distances. Note that the selective excitation of two-atom Rydberg resonances from the ground state, $2\Delta_p = E_{l,m,u}/\hbar$, requires two probe

photons and such processes are therefore second order in Ω_p , while single-atom resonances $\Delta_p = \lambda_{\pm}$ involve only one probe photon and are linear in Ω_p . An effective probe Rabi frequency for two-atom excitation is then much smaller than for a single-atom excitation, and in Fig. 2 the smaller peaks of $P_{2\text{Ry}}$ arise from a fractional (1/10) two-atom Rabi cycle, while the same amplitude and duration of the probe pulse results in several single-atom Rabi cycles leading to large and broad (sinc-shaped) peaks of $P_{1\text{Ry}}$.

Note that a small excitation probability of the antisymmetric potential curve E_{er-} seen in Fig. 2(b2) (lower left corner) is due to the dephasing γ of individual atomic coherences.

If two atoms are confined in a uniform 1D (line) or 2D (disc) volume of linear dimension L , we can average the Rydberg excitation probabilities over the interatomic distances, $\bar{P}_{1,2\text{Ry}} \equiv \int_0^L P_{1,2\text{Ry}} \varrho(R) dR$, where the corresponding probability densities for distances R are given by $\varrho_{1\text{D}}(R) = 2(L - R)/L^2$ and $\varrho_{2\text{D}}(R) = 8R/\pi L^2 [2 \arccos(R/L) - (R/2L)\sqrt{1 - (R^2/L^2)}]$. As seen in Fig. 2(c), even after averaging over a large volume, we can still discern the structure of the two-atom Rydberg excitation probabilities $\bar{P}_{2\text{Ry}}$ exhibiting an energy gap $2\Omega^{(2)} \approx 2\pi \times 0.1$ GHz between $\max E_l/2\hbar < \Delta_p < \min E_m/2\hbar$, not masked by the single-atom excitation probability $\bar{P}_{1\text{Ry}}$ within the EIT window. Such features may still persist in low-density many-atom experiments, provided the probability of having three or more atoms within a few μm distance is small compared to the two-atom probability.

We now describe a potential application of the coherent, selective excitation of the Rydberg-dimer state for quantum information processing. Assume that a pair of cold atoms, 1 and 2, are trapped at a relative distance $R_0 \approx R_m$ in an optical lattice [38,39] or by far-detuned focused laser beams [6–8]. In each atom, long-lived states $\{|s\rangle, |g\rangle\}$ represent the qubit basis states. The probe field resonantly couples the two-atom ground state $|g_1g_2\rangle$ to the bound Rydberg-dimer state with the effective two-photon Rabi frequency $\Omega_p^{(2)} \sim f\Omega_p^2/\Delta_p$, where f is the Franck-Condon overlap between the corresponding relative-coordinate wave functions [42]. Atoms in state $|s\rangle$ are decoupled from the field, while single-atom Rydberg excitations from $|g\rangle$ are suppressed by the EIT mechanism. A pulse of effective area $\theta_p = 2 \int_0^\tau \Omega_p^{(2)}(t) dt = 2\pi$ thus leads to precisely one Rabi cycle between $|g_1g_2\rangle$ and the Rydberg-dimer state, while all other initial states remain unaltered. The resulting π phase shift of $|g_1g_2\rangle$ corresponds to the two-qubit CPHASE logic gate [46,47].

We have performed simulations of the two-atom gate using realistic experimental parameters [42]. As seen in Fig. 3 (insets), sizable dephasing γ detrimentally affects the amplitude of Rabi oscillations between the two-atom

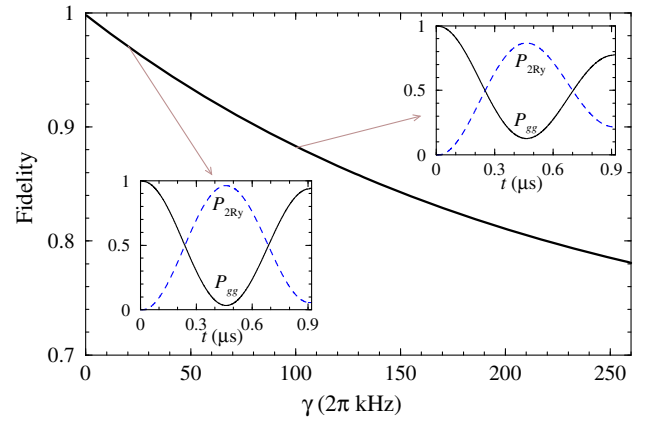


FIG. 3 (color online). Fidelity F of CPHASE gate performed on atomic qubits trapped at relative distance R_m versus dephasing rate γ . Insets show probabilities P_{gg} and $P_{2\text{Ry}}$ of two-atom ground and bound Rydberg-dimer states during one Rabi cycle, for $\gamma/(2\pi) = 20$ and 100 kHz. Probe pulse detuning is $\Delta_p = 2\pi \times 159.3$ MHz, duration $\tau = 0.9$ μs , and its single-atom Rabi frequency $\sqrt{f}\Omega_p$ is scaled by the Franck-Condon factor $f = 0.65$ [42]. Other parameters are as in Fig. 2(a).

internal-motional ground state and the lowest bound Rydberg-dimer state, and causes a leakage of population to the vibrationally excited Rydberg-dimer states [42], reducing thereby the final population $P_{gg} = \langle \hat{\sigma}_{gg}^1 \otimes \hat{\sigma}_{gg}^2 \rangle$ of state $|g_1g_2\rangle$. To quantify the performance of the gate, we apply it to the input state $|\Psi_{\text{in}}\rangle = \frac{1}{2}(|s_1\rangle + |g_1\rangle) \otimes (|s_2\rangle + |g_2\rangle)$ containing the equally weighted superposition of all two-qubit states. Ideally, the output state should be $|\Psi_{\text{out}}\rangle = \frac{1}{2}(|s_1s_2\rangle + |s_1g_2\rangle + |g_1s_2\rangle - |g_1g_2\rangle)$. In Fig. 3 we show the resulting gate fidelity $F = \langle \Psi_{\text{out}} | \hat{\rho}(\tau) | \Psi_{\text{out}} \rangle$ which is close to unity for small $\gamma \ll \Omega_p^{(2)}$, but decreases with increasing γ , as expected.

Our quantum logic gate implementation complements previous proposals [4] in several ways. For moderate interatomic separation of a few micrometers, the strong interactions are typically used for the blockade gate [2] involving resonant excitation of only one Rydberg atom [6–8], while at larger separation, both atoms can be excited and the interaction causes phase shift accumulated over time [2,9]. One can attempt to excite resonantly an antiblockaded pair of strongly interacting atoms while suppressing single-atom excitation by large detuning (equal to half the interaction energy), but then the gradient of the VDW potential amounts to a strong mechanical force between the atoms, causing motional decoherence and even excitation suppression [21]. In our implementation of a fast interaction gate, single-atom excitations are not merely suppressed by large detuning, but are almost completely canceled by destructive quantum interference, and the gate is much less vulnerable to motional decoherence since we resonantly excite the two-atom Rydberg state at a potential minimum.

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